

Faster / simpler $\rightarrow$

Helmholtz Wave Equation

$$[\nabla^2 + k_0^2 n^2(\mathbf{r})] \psi(\mathbf{r}) = 0$$

Reference full-wave formulation

Second-Order Wave ODE

$$\partial_z^2 \psi + (\nabla_\perp^2 + k_0^2 n^2(\mathbf{r})) \psi = 0$$

*Adaptive second-order
non-paraxial solve*

Forward Square-Root Propagator

$$\psi(z+\Delta z) \approx \exp(i\Delta z \sqrt{\nabla_\perp^2 + k_0^2 n^2}) \psi(z)$$

*Applied numerically with a
Lanczos/Krylov matrix-
function approximation*

WPM (Wide-angle, local n)

$$\psi(z+\Delta z) = \mathcal{F}^{-1} \left[e^{ik_z^{(m)} \Delta z} \mathcal{F}\{\psi(z)\} \right]$$

$$k_z^{(m)} = \sqrt{(k_0 n_m)^2 - k_x^2 - k_y^2}$$

*Wide-angle split-step
with local refractive index*

Angular Spectrum (Non-Paraxial)

$$\psi(z+\Delta z) = \mathcal{F}^{-1} \left[e^{ik_z \Delta z} \mathcal{F}\{\psi(z)\} \right]$$

$$k_z = \sqrt{k^2 - k_x^2 - k_y^2}$$

Wide-angle split-step in free space

Fresnel (Paraxial)

$$\psi(z+\Delta z) = \mathcal{F}^{-1} \left[e^{-i\pi\lambda\Delta z k_\perp^2} \mathcal{F}\{\psi(z)\} \right]$$

$$k_\perp^2 = k_x^2 + k_y^2$$

Paraxial split-step in free space

$\rightarrow$
choose z as propagation axis
 $\nabla^2 = \partial_z^2 + \nabla_\perp^2$

$\rightarrow$
factor into forward/backward branches
 $\partial_z \psi = i\sqrt{\nabla_\perp^2 + k_0^2 n^2} \psi$

$\rightarrow$
evaluate the square-root operator
slicewise in Fourier space
 $\sqrt{\nabla_\perp^2 + k_0^2 n^2} \rightarrow \sqrt{(k_0 n_m)^2 - k_x^2 - k_y^2}$

$\rightarrow$
homogeneous-slice limit
 $n(x, y) = \text{const. during the step}$

$\rightarrow$
small-angle Taylor expansion
 $k_z = \sqrt{k^2 - k_\perp^2} \approx k - k_\perp^2/(2k)$

$\leftarrow$ More accurate / more general